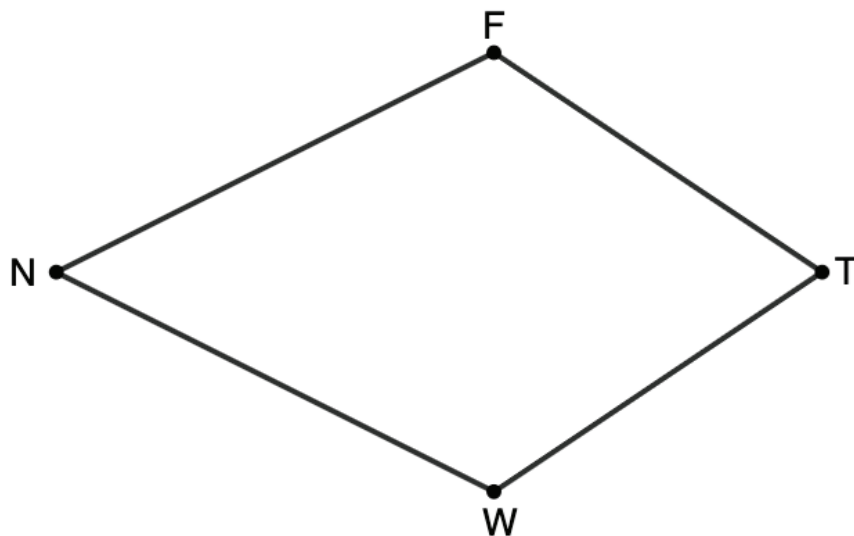




8.4.6 Wrap-Up: Cool Down

1. Kite $FTWN$ is shown. Find the value of x if $m\angle N = 53^\circ$, $m\angle T = 67^\circ$, and $m\angle F = \left(\frac{1}{2}x - 8\right)^\circ$.



Unit 8, Lesson 5: Trapezoids



Benchmarks

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



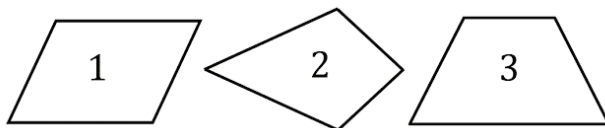
Learning Targets

- ☐ I can prove relationships and theorems about trapezoids.
- ☐ I can use relationships and theorems about trapezoids.



8.5.1 Warm-Up: Trapezoid or Trapezium?

1. Which quadrilateral is classified as a trapezoid?



There is more than one definition of a trapezoid.

Inclusive Definition	Exclusive Definition
A quadrilateral with at least one pair of parallel opposite sides.	A quadrilateral with exactly one pair of parallel opposite sides.

There is more than one definition of trapezium.

American Definition	British Definition
A quadrilateral with no parallel sides.	A quadrilateral with two sides parallel.

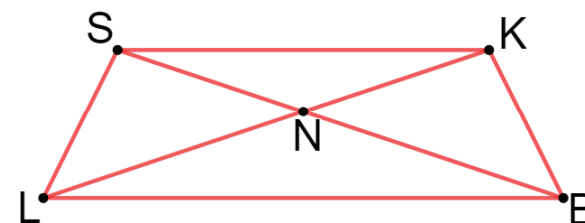
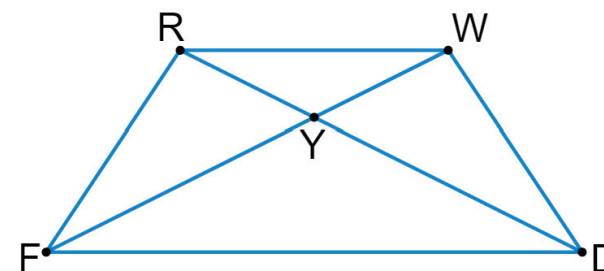
2. Use the definitions to classify quadrilaterals 1, 2, and 3. Write the numbers of the quadrilateral in the table.

Inclusive Definition of Trapezoid	Exclusive Definition of Trapezoid
American Definition of Trapezium	British Definition of Trapezium



8.5.2 Collaboration: Properties of Isosceles Trapezoids

1. $RWDF$ and $SKBL$ are both **isosceles trapezoids**. $\overline{RD}$ and $\overline{WF}$ are the diagonals of $RWDF$ and intersect at point Y . $\overline{SB}$ and $\overline{KL}$ are the diagonals of $SKBL$ and intersect at point N .



- a. Using a ruler and a protractor to measure $RWDF$ and $SKBL$, work with your partner to determine if the following conjectures about isosceles trapezoids are true or false. If false, write a new conjecture based on your measurements.

Conjecture	True or False?	If false, write a new conjecture.
All angles of an isosceles trapezoid are congruent		
Opposite sides are parallel		
Opposite angles are congruent		
Opposite sides are congruent		
Diagonals bisect each other		

- b. Compare your chart to that of another pair. Discuss any differences. Make corrections as necessary.

2. $\angle RFD$ and $\angle WDF$ in $RWDF$ are called **base angles**.
 $\angle SLB$ and $\angle KBL$ are base angles in $SKBL$.

Based on your previous measurements, make a conjecture about the base angles of an isosceles trapezoid.

3. Each isosceles trapezoid has two pairs of base angles.
- What is the other pair of base angles in trapezoid $RWDF$?
 - What is the other pair of base angles in trapezoid $SKBL$?



A **trapezoid** is a quadrilateral with *at least* one pair of parallel sides. It's a member of the following shape classes: polygons, quadrilaterals. Its sub-classes include parallelograms, rectangles, rhombuses, and squares.

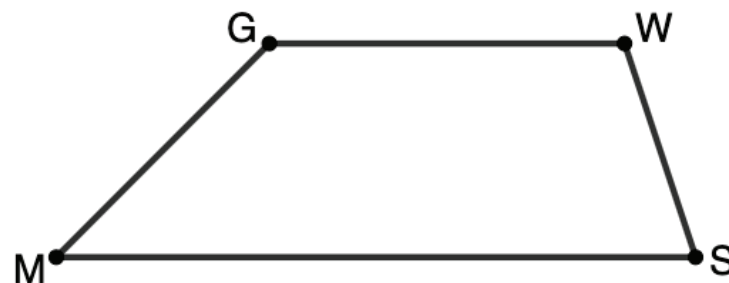
An **isosceles trapezoid** is a trapezoid with two pairs of congruent base angles.

4. Make a conjecture about the legs of an isosceles trapezoid.



8.5.3 Exploration: Parts of a Trapezoid

1. Trapezoid $GWSM$ is shown.



- a. The parallel sides of a trapezoid are called the **bases**. The trapezoid shown has one pair of bases.

Label each base red and mark them to show they are parallel on $GWSM$.

- b. The non-parallel sides of a trapezoid are called the **legs**. The trapezoid shown has two legs.

Label each leg of $GWSM$.



The **median** of a trapezoid connects the non-parallel sides at their midpoints. The median is sometimes referred to as the **midsegment**.

- c. Using your ruler, plot the midpoint of $\overline{GM}$ and the midpoint of $\overline{WS}$. Connect these points to create the median of trapezoid $GWSM$ in green. Be sure to mark the legs appropriately showing that the median is connected at the midpoints.

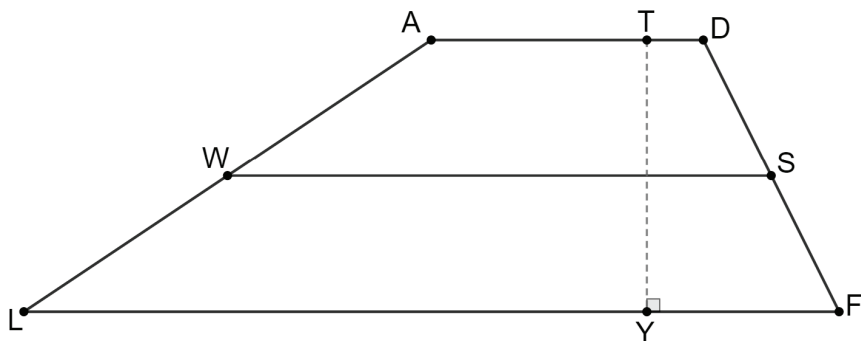


An **altitude** of a trapezoid is any segment from a point on one of the bases perpendicular to the line containing the other base.

- d. The length of the altitude is the distance between the bases.

Draw the altitude on $GWSM$. Name the altitude by naming the endpoints and mark the trapezoid to show that the altitude is perpendicular to the base.

2. Trapezoid $ADFL$ is shown with midsegment $\overline{WS}$ and altitude $\overline{TY}$.



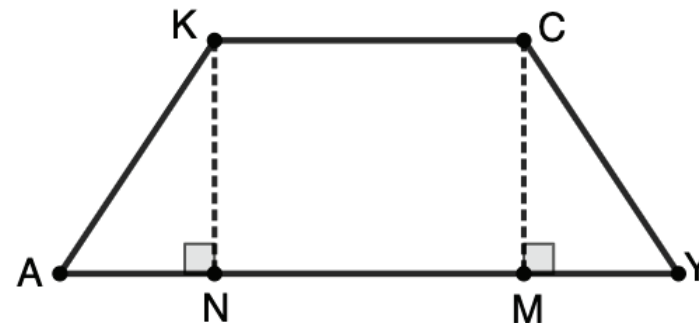
Select all of the information that is true about $ADFL$.

- | | | |
|--|--|--|
| ☐ $\overline{DF} \cong \overline{AL}$ | ☐ $m\angle LAD = m\angle FDA$ | ☐ $\overline{TY} \perp \overline{WS}$ |
| ☐ $\overline{AD} \parallel \overline{FL}$ | ☐ $\overline{WS} \parallel \overline{FL}$ | ☐ $m\angle AWS = m\angle DSW$ |
| ☐ $\overline{TY} \perp \overline{AD}$ | ☐ $\overline{DS} \cong \overline{SF}$ | ☐ $m\angle SFL = m\angle DSW$ |



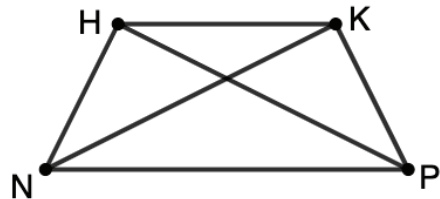
8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties

1. Trapezoid $KCYA$ is shown with altitudes $\overline{KN}$ and $\overline{CM}$.



- Trace the trapezoid onto a piece of patty paper.
- Fold the patty paper so that vertices K and C coincide.
- Describe what else you notice when you fold the patty paper.
- Work with a partner to list all of the congruent parts of trapezoid $KCYA$.
- Compare your list with that of another pair. Add any congruent parts you may have missed.
- Describe how the reflecting across the midpoint of $\overline{KC}$ shows that $KCYA$ can be classified as an isosceles trapezoid.

2. Given: $HKPN$ is an isosceles trapezoid with $\overline{HK} \parallel \overline{PN}$.
Prove: $\overline{HP} \cong \overline{KN}$

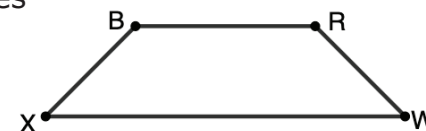


Complete the following two-column proof.

Statement	Reason
1.	1.
2. $\overline{HN} \cong \overline{KP}$	2.
3.	3. Base angles of an isosceles trapezoid are congruent
4. $\overline{NP} \cong \overline{NP}$	4.
5. $\triangle HNP \cong \triangle KPN$	5. SAS Congruence Theorem
6.	6. CPCTC

3. Given: $BRWX$ is an isosceles trapezoid and $\overline{BR} \parallel \overline{XW}$.

Prove: $\angle RBX$ and $\angle RWX$ are supplementary, $\angle BRW$ and $\angle BXW$ are supplementary.



Given that $BRWX$ is an isosceles trapezoid, then $\angle RBX \cong \angle BRW$ because _____

_____. Therefore, by the definition

of congruence, $m\angle RBX = m\angle BRW$. Since it is given that

$\overline{BR} \parallel \overline{XW}$, then $\angle RBX$ and $\angle BXW$ are _____

and $\angle$ _____ and $\angle$ _____ are supplementary

because when parallel lines are intersected by a

transversal, _____ are

supplementary. Therefore, by definition of supplementary

angles, $m\angle BRW + m\angle RWX =$ _____ and

_____ = 180° . By substitution,

$m\angle RBX + m\angle RWX = 180^\circ$ and _____ = 180° .

Therefore, by _____,

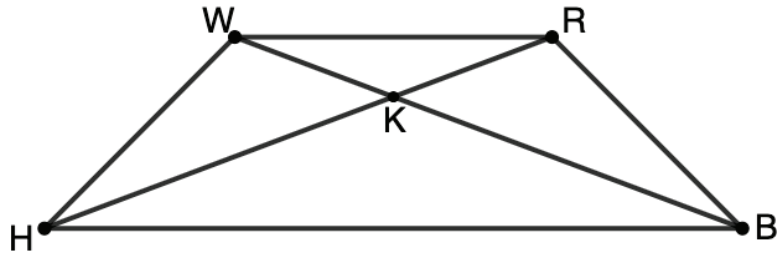
$\angle RBX$ and $\angle RWX$ are supplementary and $\angle BRW$ and $\angle BXW$

are supplementary.



8.5.5 Guided Practice: Isosceles Trapezoids

1. $WRBH$ is an isosceles trapezoid.



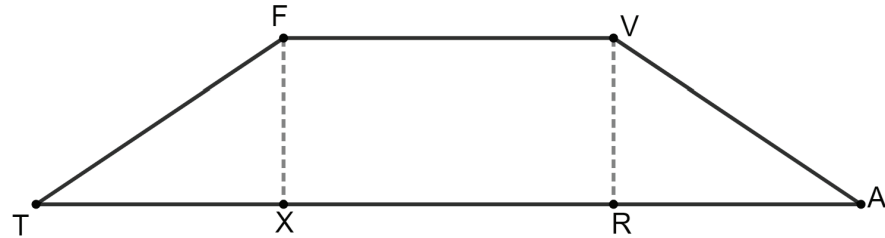
- If diagonal $WB = 18$ and diagonal $RH = 3x - 7$, find the value of x .
- Diagonals $\overline{WB}$ and $\overline{RH}$ intersect at point K . If $m\angle RWB = 20.6^\circ$, $m\angle RHW = 24.4^\circ$, and $m\angle BKH = 138.8^\circ$, what is $m\angle WRB$?

2. Isosceles trapezoid $KCYA$ has altitudes $\overline{KN}$ and $\overline{CM}$. If $KC = 6$, $AY = 30$, and $KA = 15$, what is the length of altitude $\overline{CM}$?



8.5.6 Wrap-Up: Ticket Out the Door

1. Isosceles trapezoid $FVAT$ is shown with altitudes $\overline{FX}$ and $\overline{VR}$.



If $FT = 12$, $TA = 36\sqrt{5}$, and $VR = 8$, what is the length of $\overline{FV}$?